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Exp.No- 9

Title- PRIM'S ALGORITHM

In [ ]:

```python
# Taking input from the user
n = int(input("Enter number of vertices in the graph: "))

print("Enter the adjacency matrix where each row separated by spaces:")
graph = []
for i in range(n):
    row = input().split()
    graph.append([int(x) for x in row])

def get_min_key_vertex(key, in_mst):
    return min((v for v in range(len(key)) if not in_mst[v]), key=lambda v: key[v], default=-1)

def Prim_MCST(graph):
    vertices = len(graph)
    parent = [-1] * vertices
    key = [float('inf')] * vertices
    in_mst = [False] * vertices

    key[0] = 0
    total_cost = 0

    for _ in range(vertices):
        u = min((v for v in range(vertices) if not in_mst[v]), key=lambda v: key[v], default=-1)
        if u == -1:
            break
        in_mst[u] = True

        for v in range(vertices):
            if graph[u][v] != 0 and not in_mst[v] and graph[u][v] < key[v]:
                key[v] = graph[u][v]
                parent[v] = u

    total_cost = sum(key[1:])
    return total_cost

print("===========================================================")
min_cost = Prim_MCST(graph)
print("Minimum Cost of Spanning Tree using Prim's Algorithm:", min_cost)
print("===========================================================")
```

In [ ]:

```
Enter number of vertices in the graph:   7
Enter the adjacency matrix where each row separated by spaces:
 0 28 0 0 0 10 0
 28 0 16 0 0 0 14
 0 16 0 12 0 0 0
 0 0 12 0 22 0 18
 0 0 0 22 0 25 24
 10 0 0 0 25 0 0
 0 14 0 18 24 0 0
===========================================================
Minimum Cost of Spanning Tree using Prim's Algorithm: 99
===========================================================
```